

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 04 — INTRODUCTION TO SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA)

Student Edition

Authorized ICDFA training environments only
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Lab 04: Introduction to Supervisory Control and Data Acquisition (SCADA)

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 3–4 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the purpose and architecture of SCADA systems.
- Identify the roles of PLCs/RTUs, HMI, alarms, event recording and historization.
- Start FUXA in the training SCADA VM and load a prepared HMI project.
- Connect a SCADA HMI to the OpenPLC Runtime over Modbus TCP in the isolated lab.
- Observe an automated tank-fill process and interpret operator-facing indications.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password
SCADA VM	192.168.2.10	scada	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

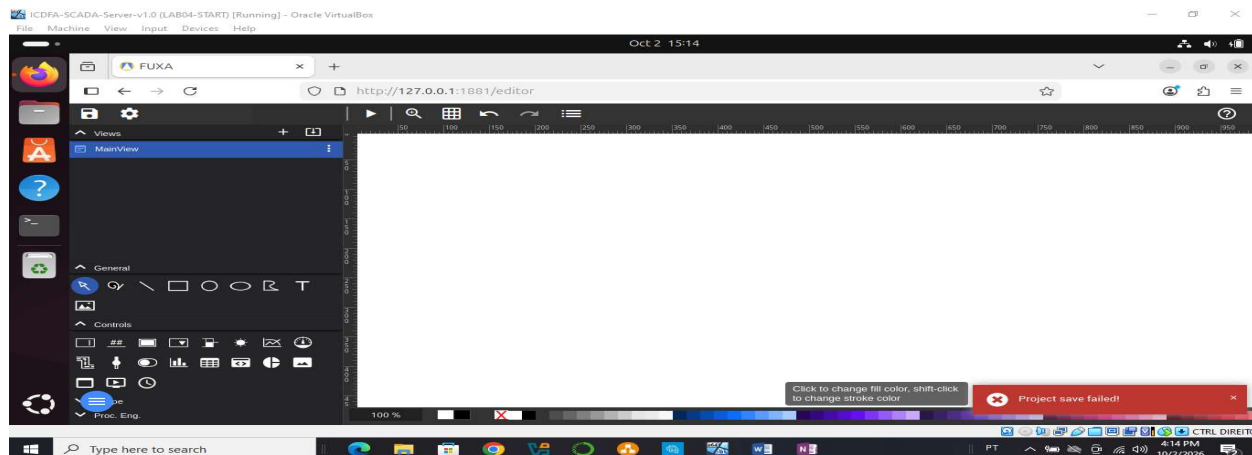
Procedure

Step 1. Import/start the ICDFA PLC and SCADA VMs using the isolated network configuration supplied by your instructor. Create LAB04-START snapshots if they are not already available.

Step 2. On the SCADA VM, sign in and open a terminal. Start FUXA using the training image command: `sudo fuxa`. Enter the SCADA account password when prompted.

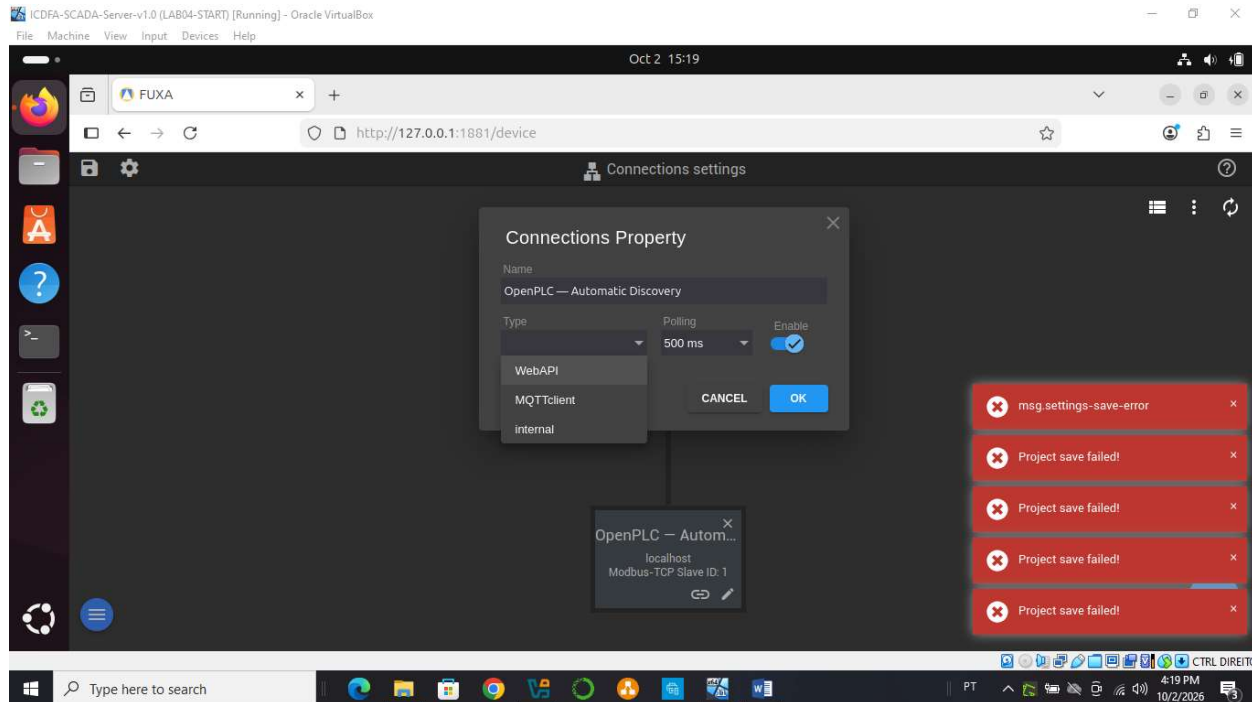
Step 3. Open Firefox and browse to <http://127.0.0.1:1881/editor> to access the local FUXA editor.

Step 4. Use the Save/Open Project function to load TankFillTest.json from the instructor-provided project location.



Note: FUXA Project save failed” repair applied successfully but the issue is not yet fixed.

Step 5. Open FUXA project settings > Connections. Edit the openplc device and set the Modbus TCP endpoint to 192.168.1.10:502. Save the connection.

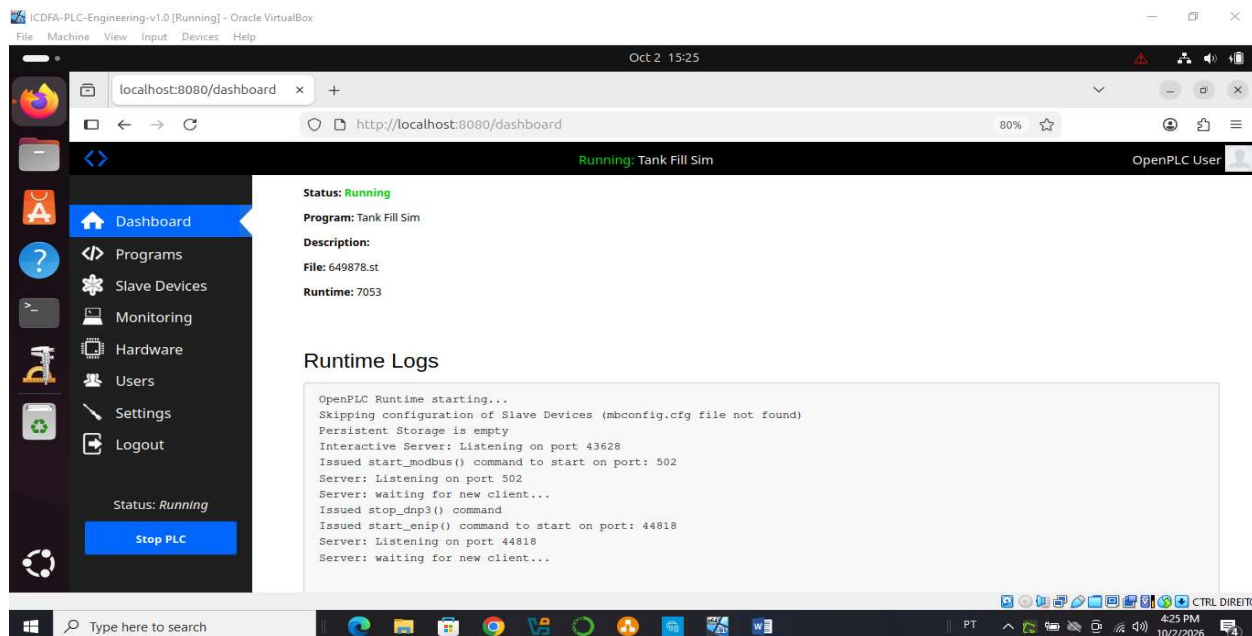


Note: Modbus setting not available

Step 6. Switch to the PLC VM. Open <http://localhost:8080/> in the local browser and sign in to OpenPLC Runtime using the credentials configured in the training image.

Step 7. Navigate to Programs, select the provided TankFillSim.st program, upload it, name it Tank Fill Sim and allow it to compile.

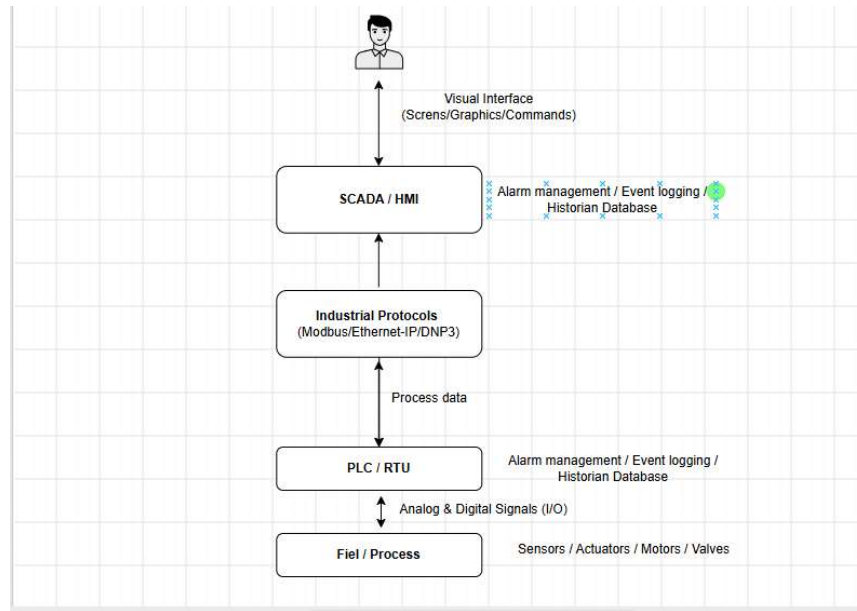
Step 8. Go to Dashboard and start the PLC. Confirm that the runtime status shows Running.



Step 9. Return to the SCADA VM and FUXA. Run/Play the HMI project.

Step 10. Observe the tank-fill simulation. Record the tank-level range, valve open/closed indication and any trend information presented to the operator.

Step 11. Create a simple architecture diagram in your report showing Field/Process -> PLC -> Industrial Protocol/Network -> SCADA/HMI -> Operator, and indicate where alarms/events/historian functions belong conceptually.



Step 12. Capture the required screenshots and stop the simulation cleanly when finished.

Explanation of the difference between control logic in the PLC and supervisory visualization in the HMI/SCADA layer.

The main difference resides on the fact that PLC control logic executes real-time machine execution, while the HMI/SCADA layer provides supervisory monitoring and data visualization.

The table below summarizes comparison among the two:

Feature	Control Logic (PLC)	Supervisory Visualization (HMI/SCADA)
Primary role	Execution: Directly drives hardware (valves, motors, actuators) based on inputs.	Supervision: Displays operational status, archives historical trends, and logs data.
Nature	Hardware-dependent: An industrial micro-computer operating at the field level.	Software-based: Applications running on touch panels, PCs, or central servers.
Speed & Timing	Deterministic real-time: Runs loops in milliseconds (1–100ms) to ensure safety and precision.	Human time-scale: Refreshes screens and polls data every few hundred milliseconds to seconds.

Required Evidence & Submission

- Screenshot showing OpenPLC Runtime in Running state.
- Screenshot showing the FUXA tank-fill HMI in operation.
- A labelled SCADA architecture diagram.
- A 150–250 word explanation of the difference between control logic in the PLC and supervisory visualization in the HMI/SCADA layer.

Recommended filename: <RegistrationNo>_LAB04_Introduction_to_Supervisory_Control_and_Data_Acquisition_SCADA.pdf

Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

Response: The primary purpose of the HMI in a SCADA System is to provide a graphical depiction of an industrial process.

2. What is the difference between an alarm, an event record and historized process data?

Response:

- ✓ Alarm - alert operators to potential problems with the process.
- ✓ Event record – is used to capture actions by both operators and automation. E.g. an event will be recorded when an operator starts a pump or turns on a system in an event log. The event log is used to recreate events to solve issues with the process.
- ✓ Historized process data – is used for trending and predictive analysis of a process.

3. Why should a SCADA operator have context rather than only raw sensor values?

Response: SCADA operators need context; raw sensor values only show a single moment, but context explains why the data matters and how to handle it safely.

4. What role does the PLC play compared with the SCADA application?

Response: PLC is a hardware component that performs direct control of a process, whereas SCADA displays that information on a graphical interface for operators to use.

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.
Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.		